

TECHNICAL DESIGN DOCUMENT

PRAKASHAN AI: IOT-BASED SOLAR SEED DRYER AND STORAGE SYSTEM

Project Identity: IoT Based Solar Seed Dryer & Storage System

Team: Agri Tech Innovation

Target Platform: Espressif ESP32-WROOM-32 Microcontroller

1. EXECUTIVE SUMMARY & OBJECTIVES

The IoT-Based Solar Seed Dryer and Storage System is an automated post-harvest agricultural solution engineered to regulate the dehydration curve of cereal and legume seeds. The system addresses two primary causes of post-harvest loss: fungal/spoilage proliferation caused by excessive moisture retention, and embryo thermal mortality caused by uncontrolled solar chamber heat.

Using an Espressif ESP32 as the central edge processor, the apparatus executes closed-loop environmental management. The controller monitors chamber ambient temperature, ambient relative humidity, internal grain moisture, and external solar irradiance. It actively drives a forced-air circulation blower and a motorized proportional exhaust flap. Telemetry is served locally on a graphical OLED display and broadcast wirelessly via an embedded HTTP web server and Bluetooth Low Energy (BLE) GATT interface.

2. VERIFIED HARDWARE SUBSYSTEMS

2.1 Central Processing Unit

The system utilizes a 32-bit dual-core ESP32 DevKit operating at 240 MHz. It provides hardware-level ADC pins for uninterrupted analog sensing, hardware PWM for servo flap positioning, dedicated I2C communication lines, and simultaneous Wi-Fi and Bluetooth wireless stacks.

2.2 Climate Monitoring

A digital DHT22 (AM2302) sensor monitors dry-bulb temperature (-40 to 80 deg C) and relative humidity (0 to 100% RH) within the drying chamber.

2.3 Dielectric Grain Moisture Sensing

A capacitive moisture sensing probe inserted directly into the grain test bed determines real-time moisture content via high-frequency dielectric permittivity measurements. This prevents the electrochemical corrosion common to resistive probes.

2.4 Signal Conditioning Modules

Dual LM393 comparator and operational amplifier boards provide adjustable analog gain, noise filtering, and threshold reference voltages for moisture and irradiance sensing.

2.5 Solar Irradiance Telemetry

A Light Dependent Resistor (LDR) divider tracks incident solar light levels to estimate passive thermal generation within the drying chamber.

2.6 Forced Convection Actuation

A 12V DC brushless blower fan mounted to the exhaust chamber duct creates directional negative pressure to extract saturated air from the seed bed.

2.7 Power Switching Driver

An optocoupled 5V single-channel relay module operates on Active-LOW logic to isolate the microcontroller from the inductive loads and electrical noise of the 12V blower fan.

2.8 Proportional Vent Servo

A high-torque micro servo motor mechanically articulates an exhaust flap between 0 degrees (fully sealed) and 90 degrees (maximum exhaust aperture).

2.9 Local User Interface

A 0.96-inch monochrome SSD1306 OLED panel (128x64 pixels, I2C interface at address 0x3C) renders real-time moisture, temperature, fan state, and system warnings.

2.10 Acoustic Safety Alarm

A 5V active piezo buzzer provides pulsed audio alerts during critical thermal events or sensor failure modes.

3. HARDWARE PIN ALLOCATION & ELECTRICAL MAPPING

All analog sensing lines are mapped to ADC1 to prevent ADC read failures during active RF transmissions.

Pin: GPIO 21

Function: I2C Data (SDA)

Connected Device: SSD1306 0.96-inch OLED Display

Signal Characteristics: 3.3V Logic, Pull-up configured

Pin: GPIO 22

Function: I2C Clock (SCL)

Connected Device: SSD1306 0.96-inch OLED Display

Signal Characteristics: 400 kHz Fast-Mode Clock

Pin: GPIO 34

Function: Analog ADC1 (Channel 6)

Connected Device: Capacitive Seed Moisture Probe

Signal Characteristics: 0 to 3.3V Analog Input (12-bit resolution)

Pin: GPIO 35

Function: Analog ADC1 (Channel 7)

Connected Device: LDR Light Sensor Comparator Module

Signal Characteristics: 0 to 3.3V Analog Input (12-bit resolution)

Pin: GPIO 32

Function: Analog ADC1 (Channel 4)

Connected Device: Battery Pack Voltage Divider (11:1 ratio)

Signal Characteristics: Scaled 0 to 35V DC input (100k/10k divider network)

Pin: GPIO 33

Function: Analog ADC1 (Channel 5)

Connected Device: Solar PV Array Voltage Divider (11:1 ratio)

Signal Characteristics: Scaled 0 to 55V DC input (100k/10k divider network)

Pin: GPIO 4

Function: Digital 1-Wire Bus

Connected Device: DHT22 Relative Humidity & Temperature Sensor

Signal Characteristics: Requires 10k ohm pull-up to 3.3V rail

Pin: GPIO 18

Function: Digital Output

Connected Device: 1-Channel Optocoupled Relay Module (Blower Fan)

Signal Characteristics: Active-LOW logic (LOW = Energized, HIGH = Off)

Pin: GPIO 19

Function: Hardware PWM Output

Connected Device: Exhaust Vent Flap Micro Servo

Signal Characteristics: 50 Hz carrier frequency, 1.0ms to 2.0ms pulse width

Pin: GPIO 23

Function: Digital Output

Connected Device: 5V Active Piezo Buzzer

Signal Characteristics: 3.3V Active-HIGH pulsed alarm

4. FINITE STATE MACHINE & CONTROL ALGORITHM

The firmware runs a non-blocking, periodic state machine with the following states:

4.1 STATE_IDLE

Default standby state upon initial power delivery or after manual user termination.

Blower Relay: De-energized (GPIO 18 = HIGH)

Vent Flap Servo: Closed (0 degrees)

Buzzer: Inactive

4.2 STATE_DRYING

Active dehydrating cycle based on crop-specific setpoints.

Blower Relay: Energized (GPIO 18 = LOW)

Vent Modulations:

If Grain Moisture > 22.0%: Flap opens to 75 degrees for high-volume moisture removal.

If Grain Moisture is between 16.0% and 22.0%: Flap adjusts to 45 degrees for balanced evaporation.

If Grain Moisture <= 16.0%: Flap restricts to 30 degrees to preserve thermal drying energy.

4.3 STATE_VENTILATING (Seed Thermal Protection)

Critical biological protection mode. Triggered when chamber temperature reaches or exceeds 42.0 deg C to prevent enzyme denaturation in cereal grains.

Blower Relay: Energized (GPIO 18 = LOW)

Vent Flap Servo: Fully opened (90 degrees) to flush excess solar heat

Visual Display: Displays overheat warning message

4.4 STATE_COMPLETED

Target moisture threshold achieved (e.g., 13.0% for paddy, 12.0% for wheat).

Blower Relay: De-energized (GPIO 18 = HIGH) to prevent over-drying and pericarp cracking

Vent Flap Servo: Positioned at 10 degrees to prevent moisture condensation while sealing against pests

Audible Indicator: Two short confirmation beeps

4.5 STATE_ALARM_ERROR

Safety shutdown executed when any sensor line disconnects or produces out-of-range telemetry (Temperature < 0 deg C, Temperature > 75 deg C, Moisture < 2%, Moisture > 50%).

Blower Relay: De-energized (GPIO 18 = HIGH)

Vent Flap Servo: Closed (0 degrees)

Buzzer: Pulsing at 2.5 Hz frequency

5. WIRELESS INTERFACES & TELEMETRY PROTOCOLS

5.1 Local Wi-Fi Access Point (SoftAP)

SSID: PRAKASHAN_AI_DRYER

Password: prakashan123

Static IP: 192.168.4.1

Subnet Mask: 255.255.255.0

5.2 Embedded REST API

Endpoint: GET <http://192.168.4.1/api/telemetry>

Response Code: 200 OK

Content-Type: application/json

Access-Control-Allow-Origin: *

Example JSON Telemetry Payload:

```
{  
  "temp": 36.5,  
  "hum": 44.0,  
  "moist": 18.2,  
  "target_moist": 13.0,  
  "solar_pct": 82,  
  "bat_v": 12.65,  
  "pv_v": 18.20,  
  "fan": 1,  
  "vent": 45,  
  "state": "DRYING",  
  "seed": "PADDY",  
  "elapsed_s": 1420  
}
```

5.3 Bluetooth Low Energy (BLE) GATT Architecture

Device Name: PRAKASHAN_AI_DRYER_BLE

Service: Nordic UART Service (NUS)

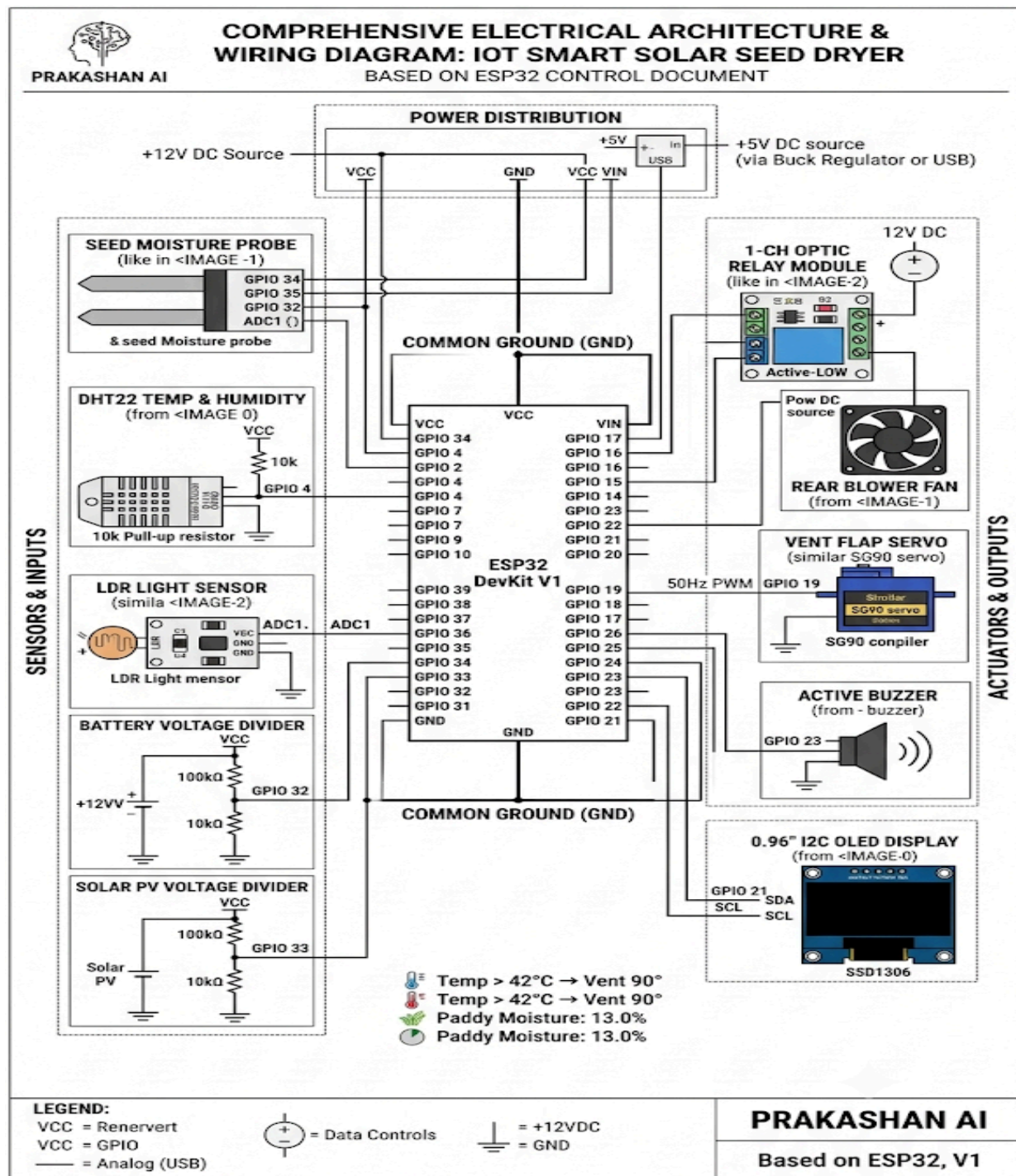
Service UUID: 6e400001-b5a3-f393-e0a9-e50e24dcca9e

TX Characteristic (Notify): 6e400003-b5a3-f393-e0a9-e50e24dcca9e

RX Characteristic (Write): 6e400002-b5a3-f393-e0a9-e50e24dcca9e

Data Fragmentation: 20-byte MTU-safe chunking with 10ms pacing between packets

6. SYSTEM ARCHITECTURAL & CIRCUIT DIAGRAM



7. ELECTRICAL SAFETY & ISOLATION REQUIREMENTS

7.1 Power Distribution Rails

The system operates two primary voltage rails:

1. 12V DC High-Power Rail: Supplies the DC blower fan through the relay common-normally open terminals.
2. 5V / 3.3V Logic Rail: Powers the ESP32, OLED, servo motor, and sensor array. Servos and relays must be supplied from a dedicated buck regulator capable of handling up to 1.5A peak stall currents without browning out the microcontroller.

7.2 Ground Reference Integrity

All external power supplies, voltage divider returns, sensor grounds, and the ESP32 ground pin must terminate at a single, common star ground bus to prevent analog ADC drift and ground loop noise.

7.3 Inductive Suppression

A freewheeling flyback diode (1N4007) must be placed in reverse parallel across the 12V DC blower fan terminals to clamp inductive voltage spikes when the relay switches off.